Author's Accepted Manuscript

image

www.elsevier.com/locate/bios

Effects of different doses of irbesartan combined with spironolactone on urinary albumin excretion rate in elderly patients with early type 2 diabetic nephropathy

Yingying Chen, Peng Liu, Xia Chen, Yanan Li, Fengmei Zhang, Yangang Wang

PII: S0002-9629(18)30052-1
DOI: https://doi.org/10.1016/j.amjms.2018.01.017
Reference: AMJMS615

To appear in: *The American Journal of the Medical Sciences*

Cite this article as: Yingying Chen, Peng Liu, Xia Chen, Yanan Li, Fengmei Zhang and Yangang Wang, Effects of different doses of irbesartan combined with spironolactone on urinary albumin excretion rate in elderly patients with early type 2 diabetic nephropathy, *The American Journal of the Medical Sciences,* doi:10.1016/j.amjms.2018.01.017

This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting galley proof before it is published in its final citable form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

ACCEPTED MANUSCRIPT

Effects of different doses of irbesartan combined with spironolactone on urinary albumin excretion rate in elderly patients with early type 2 diabetic nephropathy

Yingying Chen, MS[1*], Peng Liu, PhD[2*], Xia Chen, MS[3], Yanan Li, MS[2], Fengmei Zhang, MS[2] and Yangang Wang, PhD[4]

[1]Department of Clinical Laboratory, the People's Hospital of Laiwu, Laiwu China

[2]Department of Endocrinology, the People's Hospital of Laiwu, Laiwu China

[3]Emergency Department, the Central Hospital of Tai'an, Tai'an China

[4]Department of Endocrinology, the Affiliated Hospital of Qingdao University, Qingdao China

Corresponding authors

Peng Liu, PhD, Department of Endocrinology, the People's Hospital of Laiwu, Laiwu 271100, China, Email: sdqdlp1978@126.com; Fax: (86)0634-6279044; Telephone:0634-6279044

Yangang Wang, PhD, Department of Endocrinology, the Affiliated Hospital of Qingdao University, Qingdao 266003, China, Email: ygwang1966@126.com; Fax: (86)0532-82911740; Telephone: (86)0532-82911740

[*]Y.C and P.L contributed equally to this study.

Short title: Treatment of diabetic nephropathy in elderly with spironolactone

Conflicts of interest: The authors declare no conflicts of interest.

Role of the funding source: This study was supported by Shandong province medical and health technology development program (No.2016WS0193).

Keywords: Diabetic nephropathy; Spironolactone; Urinary albumin excretion rate

Accepted manuscript

Abstract

Background: There is a lack of research on the effect of low-dose of Angiotensin receptor blockers (ARBs) combined with spironolactone, and the effect of high-dose of ARBs alone on urinary albumin excretion rate (UAER) in elderly patients with early type 2 diabetic nephropathy (DN).

Methods: We conducted a prospective randomized, open, parallel-controlled study that included 244 elderly patients with early DN and mild-to-moderate essential hypertension. Patients were randomly divided into 4 groups: low-dose irbesartan (group A), high-dose irbesartan (group B), low-dose irbesartan combined with spironolactone

ACCEPTED MANUSCRIPT

(group C) and high-dose irbesartan combined with spironolactone (group D). Changes in UAER, serum potassium and blood pressure were compared.

Results: There were no statistical differences in the baseline characteristics among groups. Furthermore, no significant difference in blood pressure before and after treatment was found among different groups. After 72-week treatment, UAER in group D was lower compared to group A and B ($p<0.05$). Meanwhile, compared with group B, UAER in group C decreased significantly ($p<0.05$). Additionally, significantly higher serum potassium was found in group D compared to other groups ($p<0.05$). Also, group D had the highest count of patients who withdrew from the study due to hyperkalemia compared to other groups ($p<0.05$).

Conclusions: Although high-dose irbesartan combined with spironolactone has been shown to be more efficient in reducing UAER in elderly patients with early DN, this treatment is very likely to cause hyperkalemia. In addition, low-dose irbesartan combined with spironolactone has been shown to be safer and more effective in decreasing UAER compared to high-dose irbesartan.

Introduction

Diabetic nephropathy (DN) is a leading cause of end-stage renal disease, and an important risk factor for cardiovascular disease. Angiotensin receptor blockers (ARBs) are recommended as first-line agents for the treatment of DN since they can effectively reduce urinary protein and delay the decline of renal function (1). Meanwhile, many studies have suggested that high doses of ARBs have greater treatment effect compared to low dosage regimens (2). However, in some cases, aldosterone escape may reverse the beneficial effects of ARBs therapy on DN (3). Furthermore, several clinical studies have shown that the combination of spironolactone, an aldosterone receptor antagonist, and ARBs can significantly reduce the urinary protein in patients with DN (4, 5). Nevertheless, there is a lack of research on the effect of low-doses of ARBs combined with spironolactone, and the effect of high-doses of ARBs alone on renal function in elderly DN patients. The safety of different doses of ARBs combined with spironolactone in treatment of elderly patients with DN has also been very rarely reported. The following study investigated the efficacy and safety of different doses of irbesartan combined with spironolactone in elderly patients with early DN.

Accepted manuscript

ACCEPTED MANUSCRIPT

Methods

Patients

A prospective randomized, open, parallel-controlled study included 244 elderly patients with early DN and mild-to-moderate essential hypertension receiving inpatient or outpatient therapies in the Department of Endocrinology. The research project was approved by the hospital ethics committee. All patients provided informed consent.

The inclusion criteria were as follows: patients who met the Mogensen DN diagnosis and staging criteria, and the 2010 Chinese guidelines for the management of hypertension (6); age 61-75 years old; good control of blood glucose; UAER 20-199ug/min after wash out but before randomization; normal range of serum creatinine; stopping antihypertensive drugs and taking only placebo during the 2-week washout period, and blood pressure fluctuation with in the range of 140-179mmHg or 90-109mmHg for systolic (SBP) and diastolic blood pressure (DBP), respectively.

The exclusion criteria were as follows: secondary hypertension; primary renal disease; glomerular filtration rate (GFR) <45ml/minute/1.73m^2; administration of angiotensin-converting enzyme inhibitors (ACEI), ARBs, serum uric acid (SUA) synthesis-inhibiting drugs, uricosuric drugs or lipid lowering drugs for nearly 3 months; diabetic acute dysmetabolic; pregnant and lactating women; malignant tumor; severe dyslipidemia; serum potassium <3.5 mmol/L or >5.5mmol/L; and serious cardiovascular and liver and other diseases.

The termination criteria were as follows: serum potassium increased more than 5.5mmol/L; blood pressure (BP) that could not be controlled to achieve the target value; and serious adverse drug reactions that patients were not able to tolerate.

After the 2-week placebo washout period, 218 subjects were randomly divided into 4 groups: low-dose irbesartan group (group A), high-dose irbesartan group (group B), low-dose irbesartan combined with spironolactone group (group C) and high-dose irbesartan combined with spironolactone group (group D) (**Fig.1**). Randomization was performed by the block randomization method using a random number table generated by Statistics Analysis System (SAS). Group A included 53 patients (excluding 2 patients who were lost to follow-up); group B had 52 patients (excluding 1 patient with serum potassium >5.5mmol/L, and 1 patient lost to follow-up); group C had 52 patients (excluding 1 patient with serum potassium >5.5mmol/L, and 1 patient lost to follow-up); and group D included 49 patients (excluding 6 patients with serum potassium >5.5mmol/L and exit observation).

Accepted manuscript

ACCEPTED MANUSCRIPT

Determination of evaluation indicators

All patients received diabetes education, low purine diet, aspirin antiplatelet therapy, and oral hypoglycemic drugs or combined with insulin subcutaneous injection. Patients from the group A received a low dose irbesartan (150mg/day); group B received a high dose irbesartan (300mg/day); group C received a low dose irbesartan (150mg/day) combined with spironolactone (20mg/day); and patients from the group D were given a high dose irbesartan (300mg/day) combined with spironolactone (20mg/day). If BP was not under the target level after 4-week treatment, a calcium antagonists and/or beta blockers were added as a supplemental therapy. The target value of BP was 140/90mmHg. Irbesartan and spironolactone were purchased form Sanofi Hangzhou Pharmaceutical Co., LTD and Hangzhou Minsheng Pharmaceutical Co., LTD, respectively.

During the study, telephone and outpatient follow-ups were performed. A healthy lifestyle was recommended using group counseling supplemented with audiovisual and printed materials monthly. All patients were advised to give up smoking and drinking. Fasting plasma glucose (FPG) and postprandial plasma glucose (2hPG) were regularly detected, and glycated haemoglobin (HbA1c) was checked every 3 months. If FPG was >7.0mmol/l and 2hPG was >11.1mmol/l, the doses of insulin and/or oral hypoglycaemic drugs were adjusted to maintain blood glucose. Acarbose and/or prandial insulin were added if 2hPG was still >11.1mmol/l after any type of hypoglycaemic administration. The glucose target was FPG <7.0mmol/L, 2hPG 7.8-11.1mmol/L and HbA1c <7.5% (7). BP was monitored regularly. Two measurements were made, and the average recorded. To prevent any observer bias at the time of measuring BP, the nurses in charge were not familiar with the patient's information. UAER, GFR and serum uric acid (SUA) were detected and calculated once every 24 weeks, and serum potassium was detected every 4-6 weeks. If serum potassium was significantly higher, the patient was instructed to come back to the hospital for a check within 1 week. The total course of treatment was 72 weeks.

The main outcome measure was the changes of UAER in the 4 groups before and after 72-week treatment. The secondary outcome measures included changes of UAER after 24- and 48-week treatment, and changes of GFR, BP, SUA and serum potassium before and after 72-week treatment.

Oral glucose tolerance test was performed in all patients before and after treatment. All patients underwent measurements of height, weight and BP. FPG and 2hPG were determined by glucose oxidase method. High pressure liquid chromatography was used for the detection of HbA1c (Bio-Rad Inc., Hercules, CA, USA). Serum creatinine, SUA, blood lipids and serum potassium were assayed by standard enzymatic procedures on an

Accepted manuscript

ACCEPTED MANUSCRIPT

automatic biochemical analyzer (7600-020; Hitachi, Tokyo, Japan). Fasting serum insulin (FINS) was measured by radioimmunoassay (Linco Research, St. Charles, MO, USA). Urinary excretion of albumin was measured from an aliquot obtained from 3 consecutive 24-hour urine collections using a radioimmunoassay (DPC, Los Angeles, CA, USA), completed immediately at baseline and during the study. The above-mentioned indexes were detected by experienced inspection technicians, who were unaware of treatment allocation. Insulin resistance was assessed by homeostasis model assessment (HOMA-IR= FPG × FINS /22.5). Estimated GFR was calculated by CKD-EPI formula［$GFR = a\times(\text{serum creatinine}/b)^{c}\times(0.993)^{age}$)］(8).

Statistical analysis

With reference to validity and statistical requirement, we chose α=0.05 and β=0.2 (i.e., power = 80%) as the criteria for the sample size calculation. According to the corresponding reference (9), the effective percentage in UAER of the irbesartan combined with spironolactone therapy group was expected to be 25% lower compared to irbesartan treatment. The calculated results showed that a minimum of 44 cases were required for each group, with an estimated loss rate of approximately 20%. Considering that serum potassium gradually increased, a total of 244 cases were included in this study.

Data analysis was performed using SPSS 16.0 statistical software. Measurement data were analyzed by Shapiro–Wilk test, and the normal distribution data were expressed as means ± standard deviation. Differences among the groups were tested by completely randomized design 1-way analysis of variance (ANOVA), pairwise comparisons using least significant difference (LSD) test. Enumeration data were analyzed by chi-square analysis. For non-normal distributed variables, the median (P25-P75) was reported, and a Kruskal Wallis *H* test was used to examine the differences among the groups. A general linear model was used to analyze the interactions of time and group during the study. The intention-to-treat (ITT) analysis and per-protocol treated (PP) analysis were carried out on the primary outcome measure. A *p* value <0.05 was considered statistically significant.

Accepted manuscript

Results

Intergroup comparison of baseline characteristics

No statistically significant differences in baseline characteristics were found among the groups (*p*>0.05, **Table1**).

Intragroup comparison of evaluation indicators before and after treatment

ACCEPTED MANUSCRIPT

After 72-week treatment, UAER, SBP and DBP were all significantly decreased compared with pre-treatment in the 4 groups ($p<0.05$). Serum potassium was obviously higher than pre-treatment in group B, C and D ($p<0.05$). The proportion of BP achieving the target value before the addition of other antihypertensive drugs showed that group D was markedly increased compared to group A ($\chi^2=6.493$, $p<0.05$); group B increased in comparison to group A ($\chi^2=3.583$, $p>0.05$); while there was no statistical difference between group B and C ($\chi^2=1.438$, $p>0.05$, **Table2**) .

Intergroup comparison of evaluation indicators after treatment

After 72-week treatment, there were no significant differences in SBP, DBP, blood lipid and glucose among the 4 groups ($p>0.05$). UAER was significantly reduced in group C and D compared to group A ($p<0.05$). Compared with group A, UAER had a decreasing trend in group B ($p>0.05$). Furthermore, serum potassium in groups B, C and D were significantly higher compared to group A ($p<0.05$). The interaction of time and group was of no significance in SBP and DBP ($p>0.05$, **Fig.2**). The ITT analysis showed that the magnitude of UAER reduction in group D was significantly higher compared to group A and B ($p<0.05$). After 24-week treatment, UAER in group B tended to reduce compared with group C ($p>0.05$). However, after 72-week treatment, UAER reduction in group C was significantly higher compared to group B ($p<0.05$). The PP analysis results were consistent with the ITT analysis (**Table3**). Comparison of therapy-related differences showed that serum potassium was higher in group D compared to other groups ($p<0.05$). Meanwhile, lower GFR was observed in group D compared to other groups ($p<0.05$). Compared with group A and C, group B and D were more effective in reducing SUA ($p<0.05$, **Table4**). Since one of the patients in group D was treated with potassium sparing diuretic, we excluded this patient and then performed a statistical analysis, obtaining consistent results.

Adverse reactions

1 female patient (GFR 56.4ml/minute/1.73m^2) in group B, 1 male patient (GFR 58.8ml/minute/1.73m^2) in group C and 6 patients (3 male and 3 female, average GFR 63.9 ± 7.7 ml/minute/1.73m^2) in group D had serum potassium levels >5.5mmol/L. The number of the patients who could not continue to participate in the research because of hyperkalemia were higher in group D compared to group A, B and C ($\chi^2=12.390$, $p<0.05$). Serum potassium levels all gradually returned to normal range after discontinuation.

Accepted manuscript

ACCEPTED MANUSCRIPT

Discussion

In the present study, we demonstrated that the combination of irbesartan and spironolactone is more effective in lowering UAER compared to irbesartan therapy alone. After 24 weeks of treatment, higher reduction of UAER was observed in the high-dose irbesartan group; while after 72 weeks of treatment, higher reduction of UAER was found in the low-dose irbesartan combined with spironolactone group. These data indicated that the combination of low-dose irbesartan and spironolactone have a long-term effect on treating elderly patients with early DN.

Treatment with inhibitors of the renin angiotensin system (RAS), such as ARBs, has been recommended by the current guidelines as a gold-standard therapy for patients with DN. The beneficial effects of ARBs on the progression of DN are mainly attributed to the inhibition of angiotensin Ⅱ receptor activation. However, increasing evidence shows that not all patients with DN benefit from RAS blockers. Nearly 40% of these patients develop "aldosterone escape", whereby prolonged administration of ARBs treatment leads to an increase of plasma aldosterone concentration to pretreatment levels (10). Aldosterone can cause renal damage leading to glomerular and tubular fibrosis. As an aldosterone receptor antagonist, spironolactone can reduce urinary protein, thus protecting kidney function by suppressing the RAAS in local renal tissue, antagonistic inflammation and anti-fibrosis (11). Therefore, the results of this study suggest that combination of spironolactone and ARBs could be beneficial in reducing UAER and could be useful for patients where the "aldosterone escape" phenomenon is present.

Furthermore, we found that although high-dose irbesartan combined with spironolactone was more effective in decreasing UAER, it was more very likely to cause hyperkalemia; and therefore, might not be suitable for elderly patients with DN. Saito *et al* (12) also found that hyperkalemia in patients administered spironolactone was influenced by the dose, but when it was used concomitantly with ARBs, the occurrence of hyperkalemia exceeding 5.5mmol/L may increase even if the dose of spironolactone was as low as 25mg.

A number of clinical studies (4-5, 13-14) have indicated that ARBs combined with low-dose spironolactone can reduce urinary protein more effectively compared with ARBs therapy alone in patients with DN. However, our results were not completely consistent with those reported by previous studies. Firstly, we compared the effects of low-dose ARB combined with spironolactone and high-dose ARB on UAER in elderly patients with early DN. The obtained results showed that long-term effects of low-dose irbesartan combined with spironolactone on

Accepted manuscript

ACCEPTED MANUSCRIPT

reducing UAER were significantly better compared to high-dose irbesartan alone, which has not been reported in the previous researches. Secondly, two previous studies (5, 15) have shown that administration of spironolactone safely added to the renal protective benefits of treatment with maximally recommended doses of ACEI or ARBs by reducing albuminuria in DN. However, the two studies either had a short follow-up time (15) or included younger subjects (5). In the present study, we investigated the safety of long-term treatment of high-dose irbesartan combined with spironolactone in elderly DN patients. The results indicated that compared with other groups, high-dose irbesartan combined with spironolactone was more effective in reducing UAER, but it was also more likely to cause hyperkalemia. Therefore, in the present study we further optimized the treatment strategies for elderly patients with early DN.

The effects of low-dose spironolactone therapy on BP are still controversial (16, 17). In the present study, the proportion of BP achieving the target value had an increasing trend in low-dose irbesartan combined with spironolactone group compared with low-dose irbesartan group. This indicated that low-dose spironolactone may play an important role in the hypotensive. Because of the secondary outcome measure, larger sample sizes and longer follow-up time are needed to further investigate the effect of spironolactone on BP.

SUA is essential for the occurrence and development of DN (18). Liu *et al* (19) have shown that effective control of SUA may decrease UAER and exert kidney protection effects in patients with T2DM. Therefore, achievement of a target SUA is recommended by current guidelines (20). Whether irbesartan can reduce SUA still remains unclear (21, 22). In the present study, SUA could be decreased by either low-dose or high-dose irbesartan. The effect of high-dose irbesartan was more effective compared to low-dose irbesartan. Therefore, it indicated that irbesartan might decrease SUA. These findings need to be further investigated in a future study with a bigger sample size.

This study has some limitations which must be pointed out. Firstly, our study was limited by the single-center and open-label design. Secondly, plasma aldosterone was not detected before and after treatment. Thirdly, based on the IDNT study (23), high dose irbesartan has demonstrated long term benefits on doubling of serum creatinine, dialysis and death. However, due to the short follow-up period, we failed to observe the effects on “hard outcomes.” Therefore, additional prospective studies with larger population sizes and longer follow-up periods are necessary to confirm these results.

Accepted manuscript

ACCEPTED MANUSCRIPT

Conclusions

Although high-dose irbesartan combined with spironolactone has been shown to be more efficient in reducing UAER in elderly patients with early DN, this treatment is very likely to cause hyperkalemia. Low-dose irbesartan combined with spironolactone has been shown to be safer and more effective in decreasing UAER compared to high-dose irbesartan.

References

1. Chinese Diabetes Society. Guidelines for prevention and treatment of type 2 diabetes in China. Chin J Diabetes Mellitus (Article in Chinese) 2014; 7: 447-98.

2. Parving HH, Lehnert H, Brochner-Mortensen J, et al. The effect of irbesartan on the development of diabetic nephropathy in patients with type 2 diabetes. N Engl J Med 2001; 345: 870-8.

3. Schjoedt KJ, Andersen S, Rossing P, et al. Aldosterone escape during blockade of the renin-angiotensin-aldosterone system in diabetic nephropathy is associated with enhanced decline in glomerular filtration rate. Diabetologia 2004; 47: 1936-9.

4. Kato S, Maruyama S, Makino H, et al. Anti-albuminuric effects of spironolactone in patients with type 2 diabetic nephropathy: a multicenter, randomized clinical trial. Clin Exp Nephrol 2015; 19: 1098-106.

5. Mehdi UF, Adams-Huet B, Raskin P, et al. Addition of angiotensin receptor blockade or mineralocorticoid antagonism to maximal angiotensin-converting enzyme inhibition in diabetic nephropathy. J Am Soc Nephrol 2009; 20: 2641-50.

6. Writing Group of 2010 Chinese Guidelines for the Management of Hypertension. 2010 Chinese guidelines for the management of hypertension. Chin J Cardiol (Article in Chinese) 2011; 39: 579-616.

7. Chinese Society of Endocrinology. Expert consensus of HbA_{1c} control target for Chinese adult with type 2 diabetes. Chin J Endocrinol Metab (Article in Chinese), 2011; 27: 371-4.

8. Levey AS, Stevens LA, Schmid CH, et al. A new equation to estimate glomerular filtration rate. Ann Intern Med 2009; 150: 604-12.

9. Liu P, Liu R, Chen X, et al. Can levothyroxine treatment reduce urinary albumin excretion rate in patients with early type 2 diabetic nephropathy and subclinical hypothyroidism? A randomized double-blind and placebo-

Accepted manuscript

controlled study. Curr Med Res Opin 2015; 31: 2233-40.

10. Chinese Diabetes Society. Expert consensus on prevention and treatment of diabetic nephropathy (2014). Chin J Diabetes Mellitus (Article in Chinese) 2014; 6: 792-801.

11. Mavrakanas TA, Gariani K, Martin PY. Mineralocorticoid receptor blockade in addition to angiotensin converting enzyme inhibitor or angiotensin II receptor blocker treatment: an emerging paradigm in diabetic nephropathy: a systematic review. Eur J Intern Med 2014; 25: 173-6.

12. Saito M, Takada M, Hirooka K, Isobe F, Yasumura Y: Serum concentration of potassium in chronic heart failure patients administered spironolactone plus furosemide and either enalapril maleate, losartan potassium or candesartan cilexetil. J Clin Pharm Ther 2005; 30: 603-10.

13. Esteghamati A, Noshad S, Jarrah S, et al. Long-term effects of addition of mineralocorticoid receptor antagonist to angiotensin 2 receptor blocker in patients with diabetic nephropathy: a randomized clinical trial. Nephrol Dial Transplant 2013; 28: 2823-33.

14. van den Meirackera AH, Baggen RG, Paulia S, er al. Spironolactone in type 2 diabetic nephropathy: effects on proteinuria, blood pressure and renal function. J Hypertens 2006; 24: 2285-92.

15. Rossing K, Schjoedt KJ, Smidt UM, et al. Beneficial effects of adding spironolactone to recommended antihypertensive treatment in diabetic nephropathy: a randomized, double-masked, cross-over study. Diabetes Care 2005; 28: 2106-12.

16. Batterink J, Stabler SN, Tejani AM, et al. Spironolactone for hypertension. Cochrane Database Syst Rev 2010; 8: CD008169.

17. Hanazawa T, Obara T, Ogasawara K, et al. Low-dose and very low-dose spironolactone in combination therapy for essential hypertension: evaluation by self-measurement of blood pressure at home. Clin Exp Hypertens 2011; 33:427-36.

18. Chang YH, Lei CC, Lin KC, et al. Serum uric acid level as an indicator for CKD regression and progression in patients with type 2 diabetes mellitus-a 4.6-year cohort study. Diabetes Metab Res Rev 2016; 32: 557-64.

19. Liu P, Chen Y, Wang B, et al. Allopurinol treatment improves renal function in patients with type 2 diabetes and asymptomatic hyperuricemia: 3-year randomized parallel-controlled study. Clin endocrinol (Oxf) 2015; 83: 475-82.

20. Endocrine branch of the Chinese Medical Association. China expert consensus on treatment of hyperuricemia

Accepted manuscript

and gout. Chin J Endocrinol Metab (Article in Chinese) 2013; 29: 913-20.

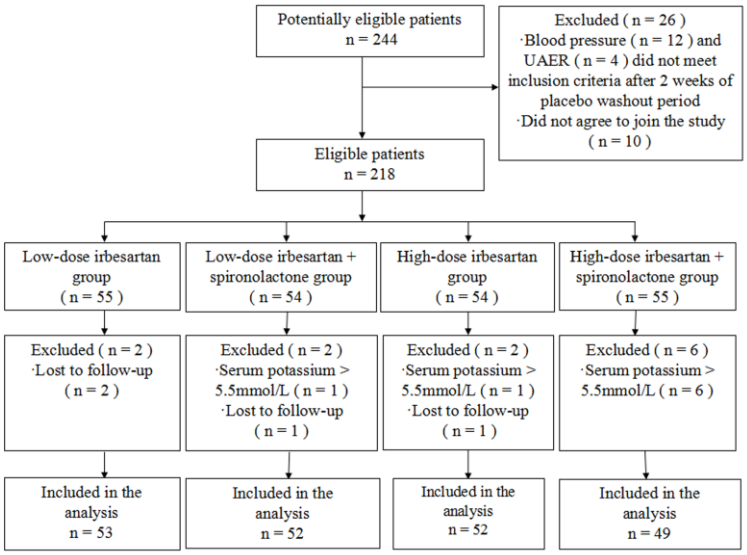
21. Nishida Y, Takahashi Y, Susa N, et al. Comparative effect of angiotensin II type I receptor blockers on serum uric acid in hypertensive patients with type 2 diabetes mellitus: a retrospective observational study. Cardiovasc Diabetol 2013; 12: 159-9.

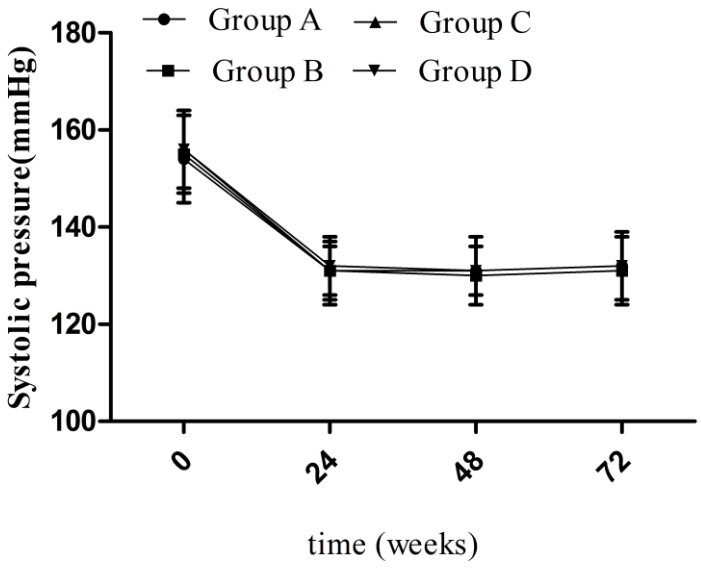
22. Wolff ML, Cruz JL, Vanderman AJ, et al. The effect of angiotensin Ⅱ receptor blockers on hyperuricemia. The Adv Chronic Dis 2015; 6: 339-46.

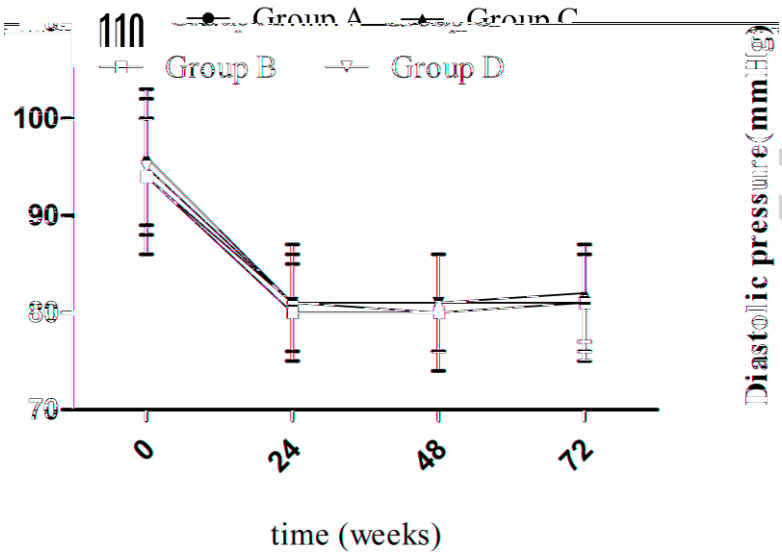
23. Lewis EJ, Hunsicker LG, Clarke WR, et al. Renoprotective effect of the angiotensin-receptor antagonist irbesartan in patients with nephropathy due to type 2 diabetes. N Engl J Med, 2001; 345: 851-60.

Figure1 Study design.

Figure2 The interaction of time and group on systolic pressure(*F*=0.267, *p*=0.983) and diastolic pressure(*F*=0.229, *p*=0.980).







Accepted Manuscript

ACCEPTED MANUSCRIPT

Table1 Baseline characteristics of the study participants

	Group A *n*=53	Group B *n*=52	Group C *n*=52	Group D *n*=49	*F* (*H* or χ^2) value	*p* value
Age, years	68 ± 4	67 ± 4	67 ± 4	67 ± 5	0.323	0.809
Gender, % (male)	28 (52.8)	27 (51.9)	25 (48.1)	26 (53.1)	0.333	0.954
Diabetes duration, years	8.0 ± 3.2	8.3 ± 3.3	7.8 ± 3.2	7.6 ± 2.7	0.481	0.696
BMI, kg/m^2	25.63 ± 1.18	25.74 ± 1.29	25.91 ± 1.30	25.69 ± 1.28	0.470	0.703
SBP, mmHg	154 ± 9	155 ± 8	156 ± 8	156 ± 8	0.572	0.634
DBP, mmHg	94 ± 6	94 ± 8	96 ± 7	95 ± 7	0.614	0.607
FPG, mmol/L	6.12 ± 0.62	6.14 ± 0.70	6.22 ± 0.66	6.17 ± 0.71	0.212	0.888
2hPG, mmol/L	9.77 ± 0.80	9.81 ± 0.73	9.95 ± 0.50	9.79 ± 0.85	0.624	0.600
HbA_1c, %	6.84 ± 0.53	6.73 ± 0.48	6.80 ± 0.52	6.77 ± 0.61	0.416	0.741
HOMA-IR	3.59 ± 0.84	3.53 ± 0.91	3.46 ± 0.83	3.51 ± 0.77	0.208	0.891
TC, mmol/L	5.36 ± 0.67	5.42 ± 0.87	5.46 ± 0.83	5.42 ± 0.75	0.130	0.942
LDL-C, mmol/L	2.79 ± 0.46	2.84 ± 0.60	2.87 ± 0.59	2.85 ± 0.63	0.205	0.893
HDL-C, mmol/L	1.03 ± 0.20	1.00 ± 0.20	1.03 ± 0.24	1.00 ± 0.18	0.493	0.687
TG, mmol/L	2.26 ± 0.60	2.31 ± 0.63	2.37 ± 0.81	2.28 ± 0.73	0.220	0.882
SUA, umol/L	443 ± 69	449 ± 77	438 ± 71	454 ± 62	0.517	0.671
UAER, ug/minute	106 (86, 142)	104 (75, 141)	104 (73, 145)	110 (73,128)	0.413	0.938
Serum creatinine, mmol/L	77 ± 9	77 ± 10	76 ± 10	76 ± 9	0.264	0.852
GFR, ml/minute/1.73m^2	79.1 ± 12.5	78.7 ± 12.7	79.3 ± 13.9	80.0 ± 13.5	0.088	0.967
Serum potassium, mmol/L	4.20 ± 0.34	4.17 ± 0.35	4.19 ± 0.34	4.13 ± 0.33	0.407	0.748
Smoking(%)	17 (32.1)	15 (28.8)	17 (32.7)	12 (24.5)	1.022	0.796
Drug use						
Oral hypoglycaemic drugs(%)	24 (45.3)	25 (48.1)	23 (44.2)	25 (51.0)	0.564	0.905
Insulin combined with oral	29 (54.7)	27 (51.9)	29 (55.8)	24 (49.0)	0.564	0.905

Accepted manuscript

ACCEPTED MANUSCRIPT

hypoglycaemic drugs(%)

BMI: Body mass index; SBP: Systolic blood pressure; DBP: Diastolic blood pressure; FPG: Fasting plasma glucose; 2hPG: 2 hour postprandial plasma glucose; HbA_1c: Glycosylated hemoglobin A1c; HOMA-IR: Homeostasis model assessment for insulin resistance; TC: Total cholesterol; LDL-C: Low density lipoprotein cholesterol; HDL-C: High density lipoprotein cholesterol; TG: Triglycerides; SUA: Serum uric acid; UAER: Urinary albumin excretion rate; GFR: Glomerular filtration rate.

Table2 Comparison of the use of antihypertensive drugs in the four groups

	Group A *n*=53	Group B *n*=52	Group C *n*=52	Group D *n*=49	χ^2 value	*p* value
Utilization of irbesartan or irbesartan combined with spironolactone alone	15 (28.3)	24 (46.2)	18 (34.6)	26 (53.1) [b]	7.927	0.048
Combined with calcium antagonist	38 (71.7)	28 (53.8)	34 (65.4)	23 (46.9) [b]	7.927	0.048
Combined with β-blockers	3 (5.7)	0 (0)	2 (3.8)	0 (0)	5.294	0.151
Combined with potassium sparing diuretics	0 (0)	0 (0)	0 (0)	1 (2.1)	3.220	0.359

vs Group A, [a]$p<0.01$, [b]$p<0.05$; *vs* Group B, [c]$p<0.01$, [d]$p<0.05$; *vs* Group C, [e]$p<0.01$, [f]$p<0.05$.

Table3 Comparison of UAER after treatment at different time points

PP analysis	Group A *n*=53	Group B *n*=52	Group C *n*=52	Group D *n*=49	*H* value	*p* value
24 weeks	-22 (-33, -15)	-28 (-43, -14)	-22 (-45, -3)	-27 (-49, -9)	3.896	0.273
48 weeks	-20 (-29, -10)	-27 (-39, -13)	-26 (-46, -14)	-31 (-56, -17) [a]	12.114	*0.007*
72 weeks	-15 (-25, -12)	-24 (-36, -12) [b]	-32 (-53, -13) [ad]	-33 (-57, -17) [ad]	23.278	*<0.001*

Accepted manuscript

ACCEPTED MANUSCRIPT

ITT analysis	Group A *n*=55	Group B *n*=54	Group C *n*=54	Group D *n*=55	*H* value	*p* value
24 weeks	-21 (-33, -14)	-27 (-42, -13)	-20 (-45, -2)	-25 (-48, -9)	2.758	0.431
48 weeks	-19 (-29, -9)	-25 (-39, -12)	-24 (-45, -13)	-28 (-54, -12) [a]	10.173	*0.017*
72 weeks	-15 (-24, -11)	-23 (-35, -12) [b]	-30 (-51, -12) [ad]	-30 (-54, -15) [ad]	20.905	*<0.001*

vs Group A, [a]$p<0.01$, [b]$p<0.05$; *vs* Group B, [c]$p<0.01$, [d]$p<0.05$; *vs* Group C, [e]$p<0.01$, [f]$p<0.05$.

Table4 Comparison of therapy-related differences among the four groups

	Group A *n*=53	Group B *n*=52	Group C *n*=52	Group D *n*=49	*F* (χ^2) value	*p* value
BMI, kg/m^2	0.02 ± 0.26	0.05 ± 0.24	0.05 ± 0.29	0.09 ± 0.25	0.576	0.631
SBP, mmHg	-23 ± 12	-24 ± 10	-24 ± 10	-24 ± 11	0.332	0.802
DBP, mmHg	-13 ± 5	-13 ± 6	-14 ± 5	-14 ± 6	0.528	0.663
FPG, mmol/L	-0.05 ± 0.90	-0.09 ± 0.98	-0.07 ± 0.95	-0.14 ± 1.03	0.083	0.969
2hPG, mmol/L	-0.03 ± 0.73	-0.07 ± 0.53	-0.08 ± 0.55	-0.10 ± 0.43	0.135	0.939
HbA_1c, %	-0.01 ± 0.40	-0.05 ± 0.42	-0.03 ± 0.38	-0.02 ± 0.36	0.104	0.958
HOMA-IR	-0.01 ± 0.29	-0.06 ± 0.36	0.05 ± 0.37	-0.07 ± 0.38	1.263	0.288
TC, mmol/L	0.13 ± 0.25	0.20 ± 0.36	0.11 ± 0.27	0.12 ± 0.24	0.983	0.402
LDL-C, mmol/L	0.10 ± 0.29	0.13 ± 0.32	0.17 ± 0.30	0.13 ± 0.31	0.422	0.737
HDL-C, mmol/L	-0.01 ± 0.04	-0.02 ± 0.05	-0.03 ± 0.07	-0.02 ± 0.07	0.659	0.578
TG, mmol/L	-0.01 ± 0.25	0.03 ± 0.36	0.07 ± 0.38	0.05 ± 0.33	0.553	0.647
SUA, umol/L	-16 ± 20	-23 ± 20 [bf]	-14 ± 17	-26 ± 18 [ae]	4.666	*0.004*
Serum creatinine, mmol/L	-1 ± 5	0 ± 6	1 ± 6	4 ± 7 [acf]	4.668	*0.004*
GFR, ml/minute/1.73m^2	-0.3 ± 4.8	-0.6 ± 5.9	-1.5 ± 6.1	-3.8 ± 5.7 [acf]	4.146	*0.007*
Serum potassium, mmol/L	0.11 ± 0.26	0.27 ± 0.19 [a]	0.29 ± 0.18 [a]	0.50 ± 0.24 [ace]	26.470	*<0.001*
Smoking(%)	11 (20.8)	9 (17.3)	10 (19.2)	8 (16.3)	0.400	0.940

vs Group A, [a]$p<0.01$, [b]$p<0.05$; *vs* Group B, [c]$p<0.01$, [d]$p<0.05$; *vs* Group C, [e]$p<0.01$, [f]$p<0.05$.

Accepted manuscript